Lesson plan · for the teacher

What is a sensor?

IoT · Level 1 · Lesson 1

Time
40 min

Materials
Three bowls or buckets of water: cold, room temperature and warm (not hot); A thermometer, if the school has one; 12 picture cards per group: sensors, displays and actuators; Board and chalk

Learning goals

- Explain that a sensor notices something in the world and turns it into a number.
- Name four sensors found at school, at home or in a phone.
- Sort everyday objects into sensors, displays and actuators.

Activity: Human sensors and real sensors: Card sort

1. Warm-up (5 min): Ask pupils to close their eyes for 30 seconds and notice everything they can: sounds, warmth, the chair, smells. Then list what they noticed and which body part noticed it.
2. Bowls test (10 min): A volunteer holds one hand in the cold bowl and one in the warm bowl for a minute, then puts both in the middle bowl. Repeat with two more volunteers.
3. Teach (8 min): Read the thermometer in the middle bowl, or explain it would give one number whoever held it. Write sensor, display and actuator on the board, each with a drawing and an example.
4. Card sort (12 min): Groups sort the 12 cards into three piles: sensors, displays and actuators. Walk around and ask each group to explain one tricky card.
5. Wrap-up (5 min): Each group shares one card they argued about. Exit ticket: each pupil writes one sensor, what it senses and where they have seen it.

Fill the bowls just before class; the warm water should feel comfortable, not hot. Draw 12 cards per group: thermometer, solar lamp sensor, soil probe, microphone, camera; TV screen, scale dial, scoreboard; pump, fan motor, buzzer, loudspeaker.

Discussion

- What did your ears, skin and nose notice just now?
- So why does each hand tell a different story?
- Does this card notice something, show something or do something?

Exit ticket

- One sensor + what it senses + where I saw it.

Try it in class tomorrow
Run the three-bowls test: Fill three bowls with cold, room-temperature and warm (not hot) water. A volunteer holds one hand in the cold bowl and one in the warm for a minute, then puts both in the middle bowl. Ask: “What does each hand say?” Then read a thermometer if you have one.

Worksheet · for pupils

What is a sensor?

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Name _____ Class _____ Date _____

1 **A pupil asks, "What is a sensor?" Which answer is best?**

- ◯ A part that notices something and turns it into a number or signal
- ◯ A part that moves or switches something on
- ◯ A screen that shows numbers to people

2 **Sensors are only found in expensive, high-tech gadgets.**

◯ True ◯ False

3 **Which of these is a sensor at work?**

◯ Wall screen ◯ Water pump ◯ Loudspeaker ◯ Probe in soil

4 **Match each sensor to what it notices.**

Glass thermometer	• • • •	How wet the soil is
Light sensor in a solar lamp	• • • •	Sound
Soil-moisture probe	• • • •	How hot or cold it is
Phone microphone	• • • •	Daylight fading

5 **A pupil lists parts of a smartphone. Which of them are sensors?**

☐ The camera ☐ The microphone ☐ The vibration motor ☐ The loudspeaker

☐ The movement sensor that turns the screen

Answers · for the teacher

What is a sensor?

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1 **A part that notices something and turns it into a number or signal**

A sensor's whole job is to notice something and turn it into a number or signal.

2 **False**

A glass thermometer is a sensor, and so is the light sensor in a low-cost solar lamp.

3 **Probe in soil**

The probe senses how wet the soil is. The pump and the loudspeaker are actuators — they act — and the screen is a display.

4 **Glass thermometer — How hot or cold it is**
Light sensor in a solar lamp — Daylight fading
Soil-moisture probe — How wet the soil is
Phone microphone — Sound

Each sensor notices one kind of thing. Naming what it senses helps pupils spot sensors everywhere — even in a lamp.

5 **The microphone; The camera; The movement sensor that turns the screen**

The microphone senses sound and turns it into a signal, the camera senses light and turns it into a picture made of numbers, and the movement sensor notices when the phone tilts or moves.

For you to think about

Which sensors could your pupils find in your school or at home this week? Name two and what each one senses.